



Knox Grammar School

Name: _____

2019 Year 11 Yearly Examination

Teacher: _____

Mathematics Advanced

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using blue or black pen only.
- Black pen is preferred.
- NESA-approved calculators may be used.
- In Questions 11-20, show relevant mathematical reasoning and/or calculations.

Examiner: Mr Willcocks

Teachers:

Mr Meli
Mr Vuletich
Mr Willcocks
Mr Cheah
Mr Lemaire
Mr Menzies
Mr Naidoo

Section I: Pages 3-6

10 marks

- Attempt Questions 1-10
- Allow about 15 minutes for this section
- Use the Multiple Choice Answer Sheet

Section II: Pages 7 - 16

90 marks

- Attempt Questions 11-20
- Allow about 2 hours 45 minutes for this section
- Answer each question in a separate writing booklet

Write your Name, your Board of Studies Student Number and your Teacher's Name on the front cover of each writing booklet.

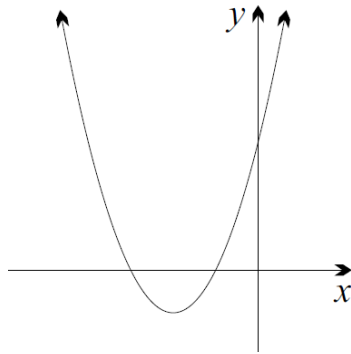
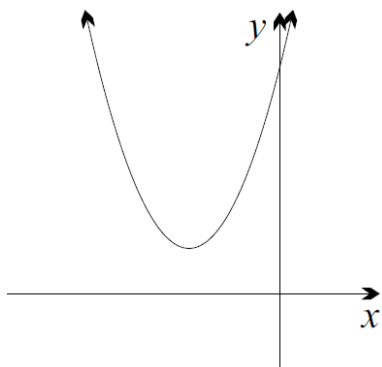
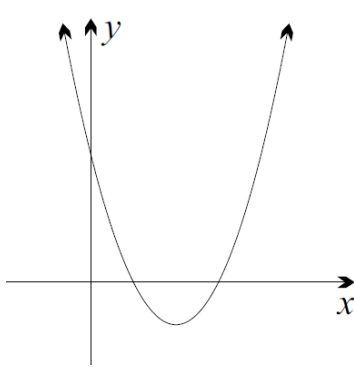
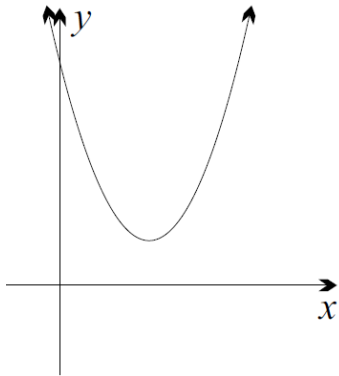
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Number of Students in Course: 202

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Section I**10 marks****Attempt Questions 1 – 10****Allow about 15 minutes for this section.**

Use the multiple choice answer sheet for questions 1 – 10.

	Marks
1 Which of the following is equal to $\frac{1}{\sqrt{5}+2\sqrt{3}}$? (A) $\frac{\sqrt{5}-2\sqrt{3}}{7}$ (B) $\frac{2\sqrt{3}+\sqrt{5}}{7}$ (C) $\frac{2\sqrt{3}-\sqrt{5}}{7}$ (D) $\frac{\sqrt{5}-2\sqrt{3}}{-7}$.	1
2 Which of the following graphs best represents $y = (x+2)^2 - 1$? (A)  (B)  (C)  (D) 	1

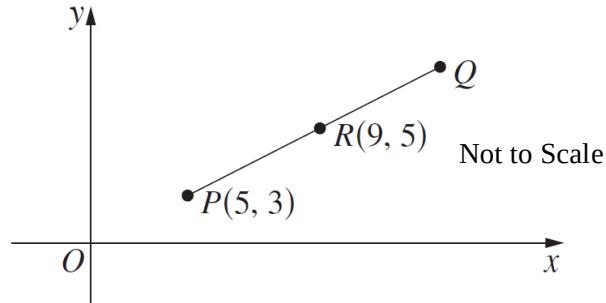
Section I continues over the page

Section I continued

Marks

- 3 The point $R(9,5)$ is the midpoint of the interval PQ , where P has coordinates $(5,3)$. What are the coordinates of Q ?

1



- (A) $(4,7)$ (B) $(7,4)$
(C) $(13,7)$ (D) $(14,8)$.

- 4 $\cos\left(\frac{-\pi}{6}\right)$ is the same as which of the following:

1

- (A) $-\cos\left(\frac{\pi}{3}\right)$ (B) $-\sin\left(\frac{\pi}{6}\right)$
(C) $\sin\left(\frac{\pi}{3}\right)$ (D) $\cos\left(\frac{2\pi}{3}\right)$.

- 5 Which of the following is the derivative of x^{n+1} with respect to x ?

1

- (A) nx^{n+1} (B) $\frac{x^{n+1}}{n+1}$
(C) $(n+1)x^{n-1}$ (D) $(n+1)x^n$.

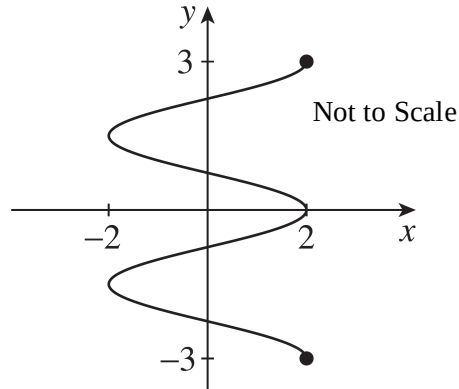
Section I continues over the page

Section I continued

Marks

- 6 Classify the graph shown below as one-to-one, one-to-many, many-to-one or many-to-many?

1



- (A) One-to-one (B) One-to-many
- (C) Many-to-one (D) Many-to-many.
- 7 Let $a = e^x$. Which expression is equal to $\log_e(a^2)$? 1
- (A) e^{2x} (B) e^{x^2}
- (C) $2x$ (D) x^2 .
- 8 Solve $\sin 2\theta = \frac{-\sqrt{3}}{2}$ in the domain $0^\circ \leq \theta \leq 180^\circ$: 1
- (A) 60° (B) No solutions
- (C) 120° (D) $120^\circ, 150^\circ$.

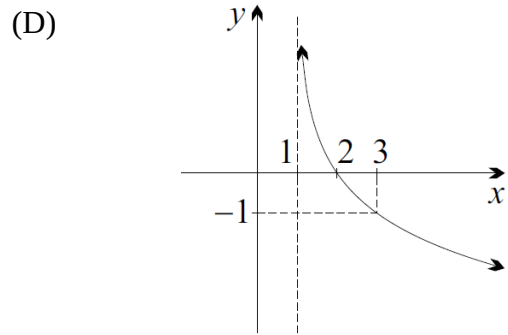
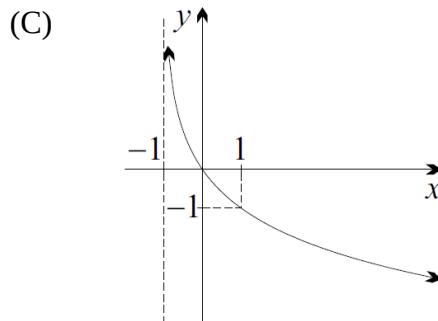
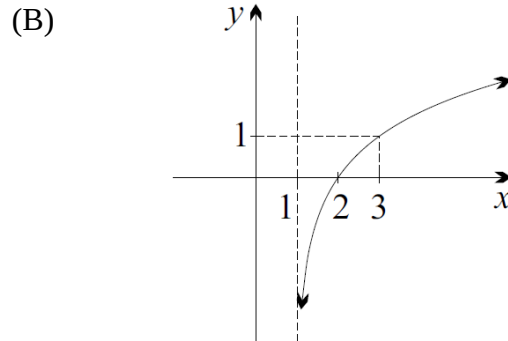
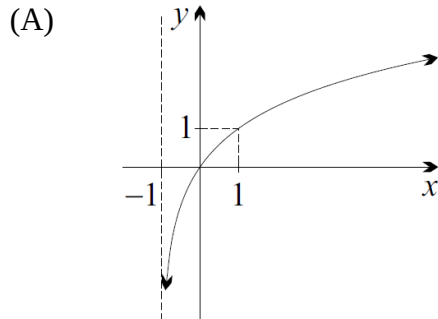
Section I continues over the page

Section I continued

Marks

9 Which of the following is the graph of $y = -\log_2(x+1)$?

1



10 Which of the following is an expression for $\frac{1-e^{2x}}{1-e^x}$?

1

(A) $1-e^x$

(B) $1+\frac{e^{2x}}{e^x}$

(C) $1+e^x$

(D) None of the above.

End of Section I

Section II**90 marks****Attempt Questions 11 – 20****Allow about 2 hours and 45 minutes for this section.**

Answer each question in the appropriate writing booklet.

Your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (9 marks)	Start a new booklet.	Marks
(a) Factorise $2x^2 - 7x - 15$.		1
(b) Find the centre and radius of the circle given by $x^2 - 6x + y^2 + 8y = 0$.		2
(c) Determine, showing all working, whether $y = 3^x - 3^{-x}$ is even, odd or neither.		2
(d) (i) Using a number plane and using half a page, sketch $g(x) = 2 - \frac{1}{x+2}$.		2
(ii) On the same diagram, sketch $y = g(-x)$, showing all asymptotes and intercepts.		1
(iii) State the range of $y = g(-x)$.		1

End of Question 11

Question 12 (9 marks)**Start a new booklet.****Marks**

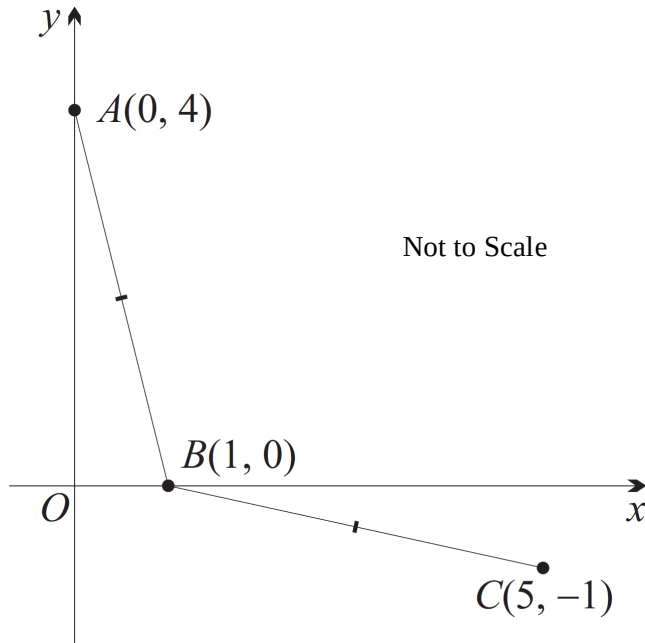
- (a) Let $f(x) = 2x + 3$ and $g(x) = 5x + b$, where b is a constant.
- (i) Find expressions for $g(f(x))$ and $f(g(x))$. **2**
- (ii) Hence find a value of b so that $g(f(x)) \equiv f(g(x))$, for all x . **1**
- (b) Differentiate $\frac{x^3 - 2x^2}{x}$ with respect to x . **1**
- (c) Given that $y = 6x\sqrt{x+1}$, find $\frac{dy}{dx}$ giving your answer as a single fraction. **2**
- (d) Consider the function $f(x) = \frac{1}{x}$.
- (i) Show that $\frac{f(x+h) - f(x)}{h} = \frac{-1}{x(x+h)}$. **1**
- (ii) Hence show that $f'(x) = \frac{-1}{x^2}$. **1**
- (iii) Describe what happens to the slope of the tangents of $f(x)$ as $x \rightarrow 0^+$ and $x \rightarrow 0^-$. **1**

End of Question 12

Question 13 (9 marks)**Start a new booklet.****Marks**

(a) Find the domain of $g(h(x))$ if $h(x) = 3x - 2$ and $g(x) = x^{\frac{1}{2}}$. **1**

(b) In the diagram below, A is $(0, 4)$, B is $(1, 0)$ and C is $(5, -1)$. The intervals AB and BC are equal in length.



- | | | |
|-------|--|----------|
| (i) | Find the gradient of AC . | 1 |
| (ii) | Find the length of AC . | 1 |
| (iii) | Show that the equation of AC is $x + y - 4 = 0$. | 1 |
| (iv) | Find the coordinates of D so that $ABCD$ is a rhombus. | 2 |
| (v) | Find the equation of BD . | 1 |
| (vi) | Find the area of the rhombus $ABCD$. | 2 |

End of Question 13

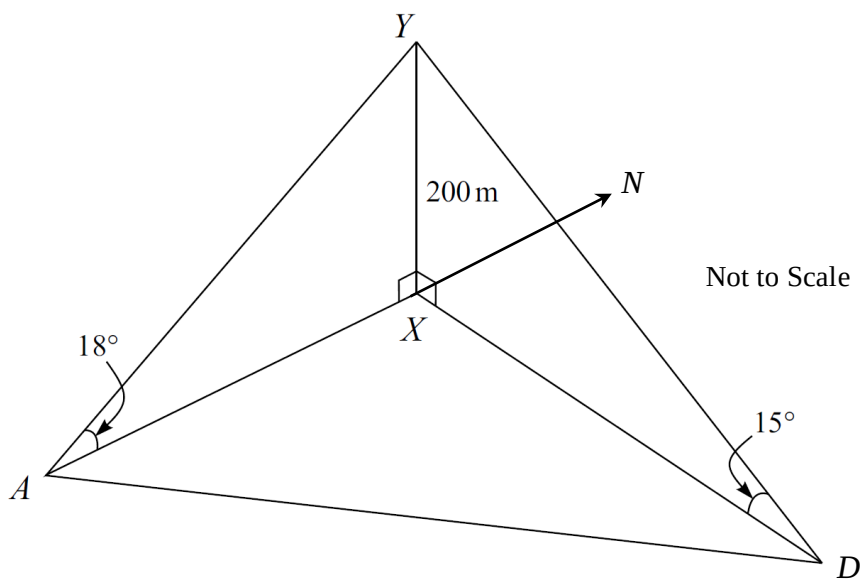
Question 14 (9 marks)

Start a new booklet.

Marks

(a) Show that $\frac{d}{dx} \left[3x^2(4x-1)^3 \right] = 6x(10x-1)(4x-1)^2$. 2

- (b) Andrew and Donna are bushwalking on flat ground. They can both see the same mountain, which has a height of 200 m. From Andrew's position A , the mountain is due north and the top of the mountain has an angle of elevation of 18° . From Donna's position D , the mountain has an angle of elevation of 15° . Let the base and the top of the mountain be the points X and Y respectively. The bearing of Donna's position from the base of the mountain is 120°T and 069°T from Andrew.



- (i) Write down the size of $\angle AXD$. 1
- (ii) Show that $XD = \frac{200}{\tan 15^\circ}$. 1
- (i) Using the sine rule, find the distance from Andrew to Donna. Give your answer correct to the nearest metre. 2
- (c) (i) Sketch $h(x) = x^2 - |2x|$, for $-3 \leq x \leq 3$. 2
- (ii) Indicate where $h(x)$ is not differentiable, giving a reason for your answer. 1

End of Question 14

Question 15 (9 marks)**Start a new booklet.****Marks**

- (a) Solve $\log_e(36 - x^2) - \log_e x = \log_e 5$ for x . **3**
- (b) A container of water, originally just below boiling point at 96°C , is placed in a fridge whose temperature is 0°C . The container is known to cool in such a way that its temperature halves every 20 minutes.
- (i) Explain why the temperature $T^\circ\text{C}$ after n hours in the fridge is given by the function $T = 96 \times \left(\frac{1}{2}\right)^{3n}$. **1**
- (ii) What is the temperature after 2 hours? **1**
- (iii) Show that the time n in hours to reach a temperature of $T^\circ\text{C}$ is given by $n = -\frac{1}{3} \log_2 \left(\frac{T}{96}\right)$. **2**
- (iv) How long does it take, correct to the nearest minute, for the temperature to fall to 1°C ? **2**

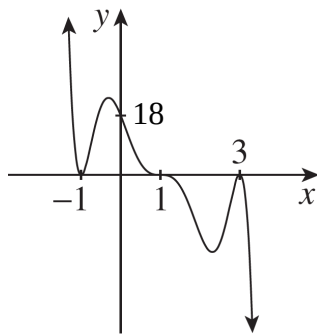
End of Question 15

Question 16 (9 marks)**Start a new booklet.****Marks**

(a) Solve $|2x-1|=5$. 2

(b) Differentiate $\frac{5x^2}{7-x}$ giving your answer in simplest form. 2

(c) The graph shown below is a polynomial which has degree 7.



Write down its equation as a product of its linear factors. 2

(d) (i) What is the domain of the function $g(x) = \frac{x^4 - x^2}{x^2 - 1}$. 2

(ii) Sketch $y = g(x)$. 1

End of Question 16

Question 17 (9 marks)**Start a new booklet.****Marks**

- (a) A bush turkey plague hit the Northern Beaches this year but was eventually brought under control. The council estimated that the bush turkey population P in hundreds, t months after 1st January, was given by $P = 15 + 14t - t^2$.
- (i) Find the bush turkey population on 1st February. **1**
- (ii) At what rate were they increasing at this time? **1**
- (iii) Calculate the average rate of increase over the first month and explain why it is different to your answer in part (ii). **2**
- (iv) In what months does the bush turkey population decrease? **2**
- (b) Solve $7^x = 6 \times 5^{x-3}$ for x , leaving your answer in terms of $\log_{10}$. **3**

End of Question 17

Question 18 (9 marks)**Start a new booklet.****Marks**

- (a) (i) Show that the equation of the tangent to $y = 1 - e^{-x}$ at the origin is $y = x$. 2
- (ii) Deduce the equation of the normal at the origin without further use of calculus. 1
- (iii) Sketch the curve, showing the points T and N where the tangent and normal respectively cut the asymptote. 2
- (b) (i) Show that $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = |\tan x|$. 2
- (ii) Hence, find the value(s) of x if $\sqrt{\frac{\operatorname{cosec}^2 x - \cot^2 x - \cos^2 x}{\cos^2 x}} = 1$ for $0^\circ \leq x \leq 180^\circ$. 2

End of Question 18

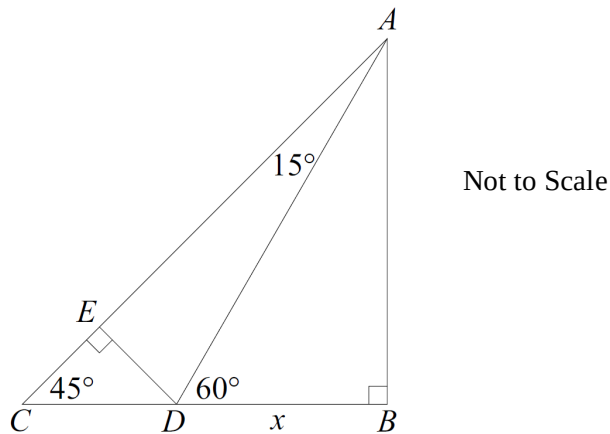
Question 19 (9 marks)

Start a new booklet.

Marks

- (a) (i) Write down the expansion for $(a+b)^2$. **1**
- (ii) Use the expansion in part (i) to show that $\left(\sqrt{6+\sqrt{11}} + \sqrt{6-\sqrt{11}}\right)^2$ is an integer. **2**
- (iii) Hence simplify $\sqrt{6+\sqrt{11}} + \sqrt{6-\sqrt{11}}$. **1**

- (b) In the diagram below, $\triangle ABC$ is a right-angled isosceles triangle. The point D is chosen on BC so that $\angle ADB = 60^\circ$, and DE is drawn perpendicular to AC . Let $BD = x$.

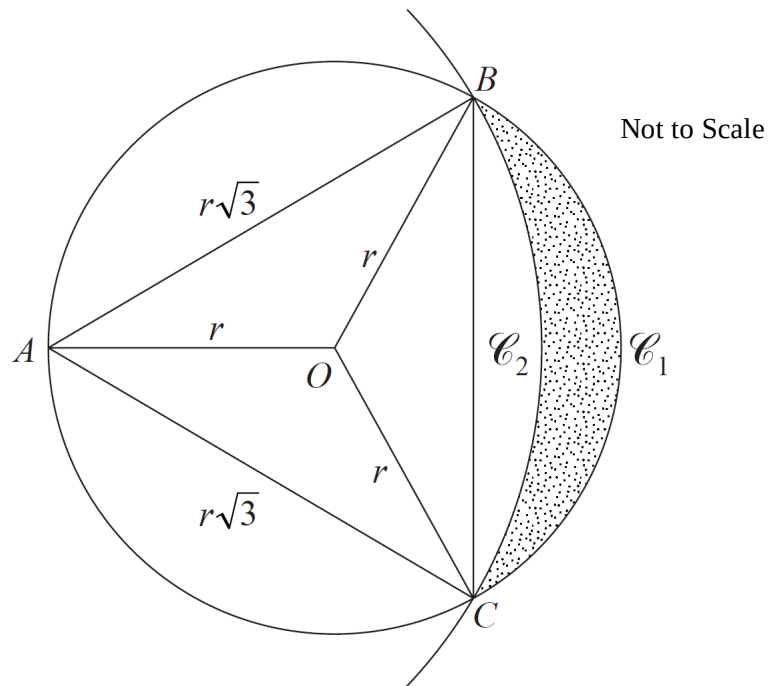


- (i) Show that $DE = \frac{(\sqrt{3}-1)x}{\sqrt{2}}$. **3**
- (ii) Hence, or otherwise, find the exact value for $\cos 15^\circ$ with a rational denominator. **2**

End of Question 19

Question 20 (9 marks)**Start a new booklet.****Marks**

- (a) Find the values of k for which $(k-2)x^2 - 2kx - 1 = 0$ has equal roots. 2
- (b) The point A lies on the circle $\mathcal{C}_1$ with centre O and radius r . The circle $\mathcal{C}_2$ with centre A and radius $r\sqrt{3}$ intersects $\mathcal{C}_1$ at the points B and C , as shown in the diagram.



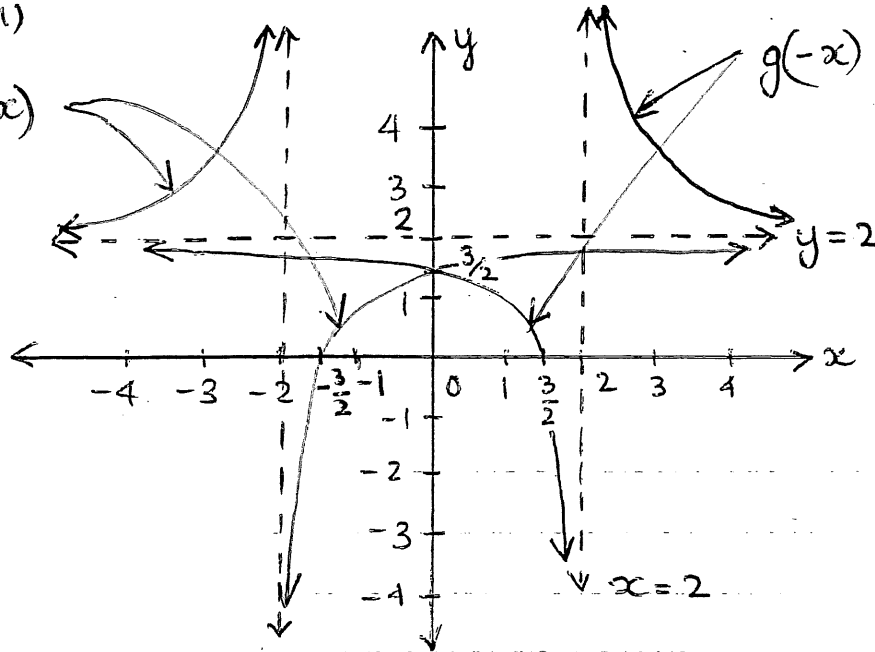
- (i) Using the cosine rule, or otherwise, find $\angle BAO$. 2
- (ii) Find the area of the sector BAC of $\mathcal{C}_2$. 1
- (iii) Find the area of the sector BOC of $\mathcal{C}_1$. 2
- (iv) Hence, or otherwise, find the area of the shaded region in terms of r . 2

End of the Examination



Multiple Choice	Suggested Solution (s)	Comments
1. $\frac{1}{\sqrt{5}+2\sqrt{3}} \times \frac{\sqrt{5}-2\sqrt{3}}{\sqrt{5}-2\sqrt{3}}$ $= \frac{\sqrt{5}-2\sqrt{3}}{5-12}$ $= \frac{2\sqrt{3}-\sqrt{5}}{7} \text{ or } \frac{\sqrt{5}-2\sqrt{3}}{-7}$ $\Rightarrow \text{C/D}$	6. For at least one number in the domain there is more than one number in range, every number in range there is only one number in domain $\Rightarrow \text{B}$	
2. $y = (x+2)^2 - 1$ $\uparrow$ move down 1 shift left 2 positive quadratic $\Rightarrow \text{A}$	7. $\log_e(e^{2x})^2$ $= \log_e e^{2x}$ $= 2x \Rightarrow \text{C}$	
3. let $Q(x, y)$ $\frac{5+x}{2} = 9 \therefore x = 13$ $\Rightarrow \text{C}$	8. $2\theta = 180 + 60^\circ$ $360 - 60^\circ$ $\therefore \theta = 120^\circ, 150^\circ$ $\Rightarrow \text{D}$	
4. $\cos\left(-\frac{\pi}{6}\right)$ $= \cos\left(\frac{\pi}{6}\right)$ $= \sin\left(\frac{\pi}{2} - \frac{\pi}{6}\right)$ $= \sin\left(\frac{\pi}{3}\right) \Rightarrow \text{C}$	9. $y = -\log_2(x+1)$ $\uparrow$ flip x $\uparrow$ shift left $\Rightarrow \text{C}$	
5. $\frac{d}{dx}(x^{n+1})$ $= (n+1)x^{n+1-1}$ $= (n+1)x^n \Rightarrow \text{D}$	10. $\frac{(1+e^x)(1-e^x)}{(1-e^x)}$ $= 1+e^x \Rightarrow \text{C}$	
<u>Summary</u> 1. C/D 2. A 3. C 4. C 5. D 6. B 7. C 8. D 9. C 10. C		



Question. 11	Suggested Solution (s)	Comments
(a) $(2x+3)(x-5) \checkmark$		
(b) $x^2 - 6x + 9 + y^2 + 8y + 16 = 25$ $(x-3)^2 + (y+4)^2 = 5^2$ centre $(3, -4)$ radius $= 5 \checkmark$		
(c) $f(-x) = 3^{-x} - 3^{-(-x)}$ $= -(3^x - 3^{-x}) \checkmark$ $= -f(x)$ $\therefore$ odd $\checkmark$		
(d)(i)(ii)		
(i) max 2 marks - 1 no intercepts - 1 no asymptotes	(ii) can be on separate graphs - no penalty must show reflection about y-axis,	
(iii) range $y \in \mathbb{R}, y \neq 2$		



Question 12	Suggested Solution (s)	Comments
	(a)(i) $g(f(x)) = 5(2x+3)+b$ ✓ $= 10x + 15 + b$ $f(g(x)) = 2(5x+b)+3$ ✓ $= 10x + 3 + 2b$ (ii) $15+b = 3+2b$ $b = 12$ ✓ (b) $= \frac{d}{dx}(x^2 - 2x)$ $= 2x - 2$ ✓ (c) $u = 6x$ $v = (x+1)^{\frac{1}{2}}$ $u' = 6$ $v' = \frac{1}{2}(x+1)^{-\frac{1}{2}}$ ✓ $= 6(x+1)^{\frac{1}{2}} + 3x(x+1)^{-\frac{1}{2}}$ ✓ $= \frac{3(2x+1)}{\sqrt{x+1}}$ or $\frac{3(2x+1)\sqrt{x+1}}{x+1}$ ✓ either (ii) $f'(x) =$ the limit of $h \rightarrow 0$ of $\frac{f(x+h) - f(x)}{h}$ so that the gradient of the secant becomes the gradient of the tangent at point x. (ii) $f'(x) = h \rightarrow 0$ of $\frac{f(x+h) - f(x)}{h}$ $= \frac{-1}{x(x+0)}$ $= \frac{-1}{x^2}$ student must convey that $h \rightarrow 0$ (iii) $f'(x) \rightarrow -\infty$ ✓	1 mark either line



Question . 13

Suggested Solution (s)

Comments

$$(a) \quad g(h(x)) = \sqrt{3x-2}$$

$$\text{domain } 3x-2 \geq 0$$

$$x \geq \frac{2}{3} \quad \checkmark$$

must have
"greater than or
equal to."

$$(b) \quad (i) \quad m_{AC} = \frac{-1-4}{5-0} \\ = -1 \quad \checkmark$$

$$(ii) \quad d_{AC} = \sqrt{(-1-4)^2 + (5-0)^2} \\ = \sqrt{50} \\ = 5\sqrt{2} \quad \left. \vphantom{\begin{matrix} d_{AC} \\ \sqrt{50} \end{matrix}} \right\} \text{either } \checkmark$$

$$(iii) \quad y-4 = -1(x-0) \quad \checkmark \\ y+x-4=0$$

$$(iv) \quad \text{let } D(x, y)$$

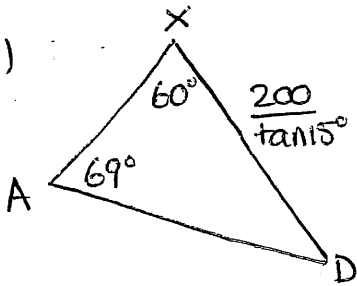
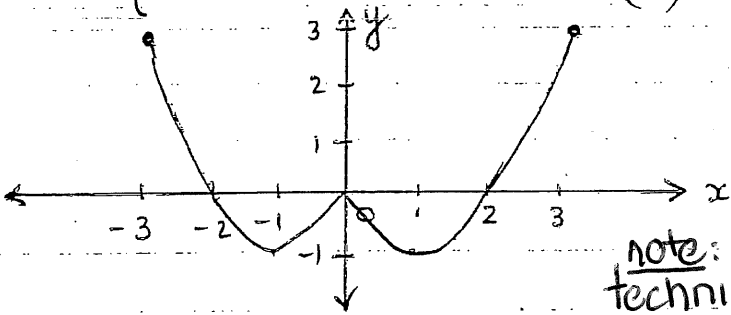
$$\frac{1+x}{2} = \frac{5}{2} \quad \text{and} \quad \frac{0+y}{2} = \frac{3}{2}$$

$$\therefore x=4 \quad \checkmark, \quad y=3 \quad \checkmark$$

$$(v) \quad y-0 = 1(x-1) \\ x-y-1=0 \quad \checkmark \quad \dots \text{or equivalent.}$$

$$(vi) \quad d_{BD} = \sqrt{(3-0)^2 + (4-1)^2} = 2 \times \left(\frac{1}{2} \times 5 \times 5 - \right. \\ = \sqrt{18} \\ = 3\sqrt{2} \quad \checkmark \quad \left. \vphantom{\begin{matrix} d_{BD} \\ \sqrt{18} \end{matrix}} \right) \quad \text{or} \quad \frac{1}{2} \times 4 \times 1 = \checkmark \\ A_{ABCD} = \frac{1}{2} \times 3\sqrt{2} \times 5\sqrt{2} \quad \frac{1}{2} (1+5) \\ = 15u^2 \quad \checkmark \quad = 15u^2 \quad \checkmark$$



Question . 14	Suggested Solution (s)	Comments
	(a) let $u = 3x^2$, $v = (4x-1)^3$ $u' = 6x$, $v' = 12(4x-1)^2$ ✓ both $\frac{d}{dx} (3x^2(4x-1)^3) = 6x(4x-1)^3 + 36x^2(4x-1)^2$ $= 6x(4x-1)^2(4x-1+6x)$ $= 6x(10x-1)(4x-1)^2$ must show factoring line for 2 marks (b)(i) $\angle AXD = 180^\circ - 120^\circ$ $= 60^\circ$ ✓ (ii) $\tan 15^\circ = \frac{200}{XD}$ ✓ $XD = \frac{200}{\tan 15^\circ}$ or $200 \cot 15^\circ$ (iii)  $\frac{AD}{\sin 60^\circ} = \frac{200}{\tan 15^\circ} \div \sin 69^\circ$ ✓ $AD = \frac{100\sqrt{3}}{\tan 15^\circ \sin 69^\circ}$ $AD = 692.399...$ $AD = 692 \text{ m (nearest metre)}$ ✓ (c)(i) $h(x) = \begin{cases} x^2 - 2x, & x \geq 0 \\ x^2 + 2x, & x < 0 \end{cases}$ $h(3) = 3$ $h(-3) = 3$  note: technically <u>not</u> differentiable at end points also $\Rightarrow$ no marks allocated here. (ii) as $x \rightarrow 0^-$, $m > 0$ $x \rightarrow 0^+$, $m < 0$ <u>not</u> smooth at $x=0$ (ie cusp) ✓	



Question 15

Suggested Solution (s)

Comments

$$(a) \log_e \left(\frac{36 - x^2}{x} \right) = \log_e 5$$

$$\therefore 36 - x^2 = 5x$$

$$x^2 + 5x - 36 = 0$$

$$(x + 9)(x - 4) = 0 \quad \checkmark$$

$$\therefore x = 4 \quad \checkmark$$

$$\therefore 0 < x < 6 \quad \text{or}$$

demonstrated that for $\log x$, $x > 0$

(b) (i) starts at 96°C and halves every 20 mins. Therefore $3n$ is the number of 20 min periods in n hours. $\checkmark$

$$(ii) T = 96 \times \left(\frac{1}{2}\right)^6$$

$$= 1.5^\circ\text{C} \quad \checkmark$$

$$(iii) \frac{T}{96} = \left(\frac{1}{2}\right)^{3n}$$

$$\log_{\frac{T}{96}} = 3n \log_2 \left(\frac{1}{2}\right) \quad \checkmark$$

$$n = \frac{\frac{1}{3} \times \log_{\frac{T}{96}}}{-1} \quad \checkmark \quad \text{either } \log_2 2^{-1} \text{ or } -1$$

this needs to be clear for second mark.

$$n = -\frac{1}{3} \log_2 \left(\frac{T}{96}\right)$$

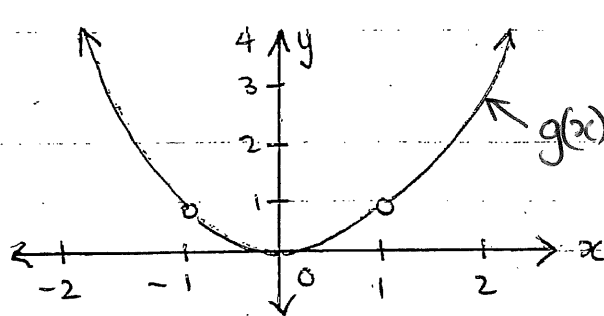
$$(iv) n = \frac{\log 96 - \log 1}{3 \log 2}$$

$$= 2.194 \dots \text{ hours} \quad \checkmark \quad \text{change of base}$$

$$= 2^\circ 11' 41.96''$$

$$= 2 \text{ hours } 12 \text{ mins} \quad \checkmark$$



Question. 16	Suggested Solution (s)	Comments
	(a) $2x - 1 = 5$ and $2x - 1 = -5$ $2x = 6$ $2x = -4$ $x = 3$ ✓ $x = -2$ ✓ (b) $u = 5x^2$ $v = 7 - x$ ✓ $= \frac{70x - 5x^2}{(7-x)^2}$ $u' = 10x$ $v' = -1$ $\frac{10x(7-x) + 5x^2}{(7-x)^2}$ ✓ $= \frac{5x(14-x)}{(7-x)^2}$ (c) $y = a(x+1)^2(x-1)^3(x-3)^2$ ✓ sub in $(0, 18)$ to find a $18 = -9a$ $\therefore a = -2$ and $y = -2(x+1)^2(x-1)^3(x-3)^2$ ✓ (d)(i) $g(x) = \frac{x^2(x^2-1)}{x^2-1}$ domain: $x \in \mathbb{R}, x \neq \pm 1$ ✓ x^2 ✓, $x \neq \pm 1$ (ii)  ✓ ○ no holes	



Question 17

Suggested Solution (s)

Comments

$$(a)(i) \quad P = 15 + 14(t) - t^2$$

$$P = 28$$

$\therefore$ population 1st Feb = 2800 ✓

$$(ii) \quad \frac{dP}{dt} = 14 - 2t$$

at $t = 1 \quad \frac{dP}{dt} = 12 \quad \therefore$ increasing at 1200/month. ✓

$$(iii) \quad \text{increase is } 2800 - 1500 = 1300$$

Average rate is 1300/month ✓

the rate in part (ii) is smaller

because it is instantaneous and slowing down over time. ✓

$$(iv) \quad \frac{dP}{dt} = 0 \quad 2t = 14 \quad (15-t)(1+t) = 0$$

$$t = 7 \quad \text{1 mark for } 7 \neq 15$$

$7 < t \leq 15$ or Aug - March (following year) ✓

$$(b) \quad \log_{10} 7^x = \log_{10} 6 + \log_{10} 5^{x-3} \quad \checkmark$$

$$x \log_{10} 7 = \log_{10} 6 + (x-3) \log_{10} 5$$

$$x (\log_{10} 7 - \log_{10} 5) = \log_{10} 6 - \log_{10} 5^3 \quad \checkmark \quad \text{collecting and factoring } x$$

$$x = \frac{\log_{10} \frac{6}{125}}{\log_{10} \frac{7}{5}} \quad \text{or} \quad \frac{\log_{10} 6 - \log_{10} 125}{\log_{10} 7 - \log_{10} 5} \quad \checkmark$$



Question. 18

Suggested Solution (s)

Comments

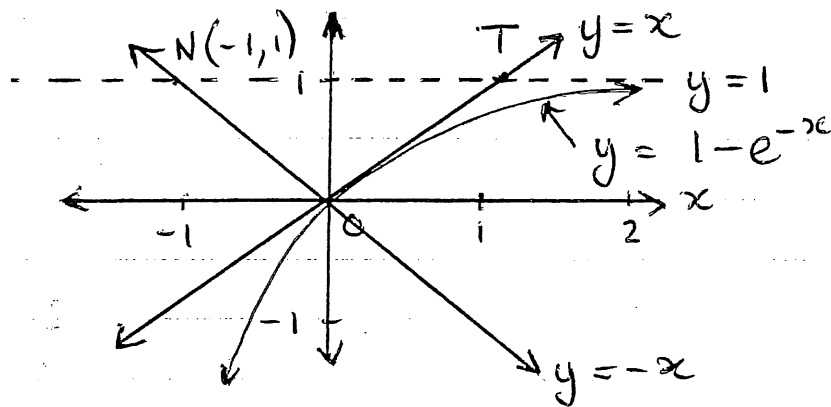
(a) (i) $\frac{dy}{dx} = e^{-x}$ ✓

at $x=0$ $\frac{dy}{dx} = 1$ ✓, $y=0$

$\therefore y = x$

(ii) normal: $y = -x$ ✓

(iii)



(b) (i)

LHS = $\sqrt{\frac{1 + \cancel{\cot^2 x} - \cancel{\cot^2 x} - \cos^2 x}{\cos^2 x}}$ ✓

= $\sqrt{\frac{1 - \cos^2 x}{\cos^2 x}}$

= $\sqrt{\frac{\sin^2 x}{\cos^2 x}}$ ✓

= $\sqrt{\tan^2 x}$

= $|\tan x|$

(ii) $|\tan x| = 1$

$\tan x = \pm 1$

$\therefore x = 45^\circ, 135^\circ$ 1 each.



Question 19	Suggested Solution (s)	Comments
(a)(i) $a^2 + 2ab + b^2$ ✓	(ii) let $a = \sqrt{6+\sqrt{11}}$, $b = \sqrt{6-\sqrt{11}}$	
$(a+b)^2 = 6 + \sqrt{11} + 2\sqrt{36-11} + 6 - \sqrt{11}$ ✓ $= 12 + 10$ $= 22 \in \mathbb{Z}$ ✓	(iii) $= a+b$ $= \sqrt{22}$ ✓	
(b)(i) $\tan 60^\circ = \frac{AB}{x}$	$AB = \sqrt{3}x$ $AB = BC$ (isosceles triangle)	
$\therefore CD = \sqrt{3}x - x$ ✓ $= (\sqrt{3}-1)x$	$CE = DE$ (isosceles triangle)	
$\therefore 2DE^2 = (\sqrt{3}-1)^2 x^2$ ✓	$DE = \frac{(\sqrt{3}-1)x}{\sqrt{2}}$	
(ii) $AD^2 = x^2 + (\sqrt{3}x)^2$	$AD = 2x$	
$AE^2 = (2x)^2 - \frac{(\sqrt{3}-1)^2 x^2}{2}$ ✓	$AE^2 = \frac{8x^2}{2} - \frac{(3-2\sqrt{3}+1)x^2}{2}$	
$AE = \frac{\sqrt{8-4+2\sqrt{3}}}{\sqrt{2}} x$	$AE = \frac{\sqrt{2} \sqrt{4+2\sqrt{3}}}{2} x$	note: $\frac{\sqrt{4+2\sqrt{3}}}{2\sqrt{2}}$ 2 marks only.
$\cos 15^\circ = \frac{AE}{AD} = \frac{\sqrt{8+4\sqrt{3}}}{4}$ ✓✓		



Question. 20	Suggested Solution (s)	Comments
	(a) $\Delta = b^2 - 4ac$ $0 = 4k^2 - 4(-1)(k-2)$ $\therefore 4k^2 + 4k - 8 = 0 \checkmark$ $k^2 + k - 2 = 0$ $(k+2)(k-1) = 0$ $\therefore k = -2 \text{ or } k = 1 \checkmark$ (b) (i) $\cos \angle BAO = \frac{r^2 + (r\sqrt{3})^2 - r^2}{2(r)(r\sqrt{3})}$ $= \frac{3\cancel{r^2} \times \sqrt{3}}{2\sqrt{3}\cancel{r^2} \times \sqrt{3}} \checkmark$ $= \frac{\sqrt{3}}{2}$ $\therefore \angle BAO = \cos^{-1} \frac{\sqrt{3}}{2} \checkmark$ $= \frac{\pi}{6}$ (ii) $A_1 = \frac{1}{2}(r\sqrt{3})^2 \frac{\pi}{3}$ $\angle BAO = \angle CAO$ (isosceles triangles) $A_1 = \frac{\pi r^2}{2} \checkmark$ $\therefore \angle CAO = \frac{\pi}{3}$ (iii) $\angle BOA = \pi - \frac{\pi}{3}$ (L sum Δ) $= \angle AOC$ $\angle BOC = 2\pi - \frac{4\pi}{3}$ $= \frac{2\pi}{3} \checkmark$ $A_2 = \frac{1}{2} r^2 \left(\frac{2\pi}{3} \right)$ $= \frac{\pi r^2}{3} \checkmark$ (iv) $A_{\Delta AOB} = A_{\Delta AOC} = A_{\Delta BOC} = \frac{1}{2} r^2 \sin \left(\frac{2\pi}{3} \right)$ $= \frac{1}{2} r^2 \frac{\sqrt{3}}{2} = \frac{\sqrt{3}}{4} r^2 \checkmark$ $A_{\text{shaded region}} = \frac{2\sqrt{3}r^2}{4} - \frac{\pi r^2}{2} + \frac{\pi r^2}{3}$ $= \left(\frac{\sqrt{3}}{2} - \frac{\pi}{2} + \frac{\pi}{3} \right) r^2 = \left(\frac{\sqrt{3}}{2} - \frac{\pi}{6} \right) r^2 \checkmark \text{ or } 0.34r^2$	